Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
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Practical Answer Submission

Practical Sheet No: 08

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The use of Shared memory in C language.

Activity 1

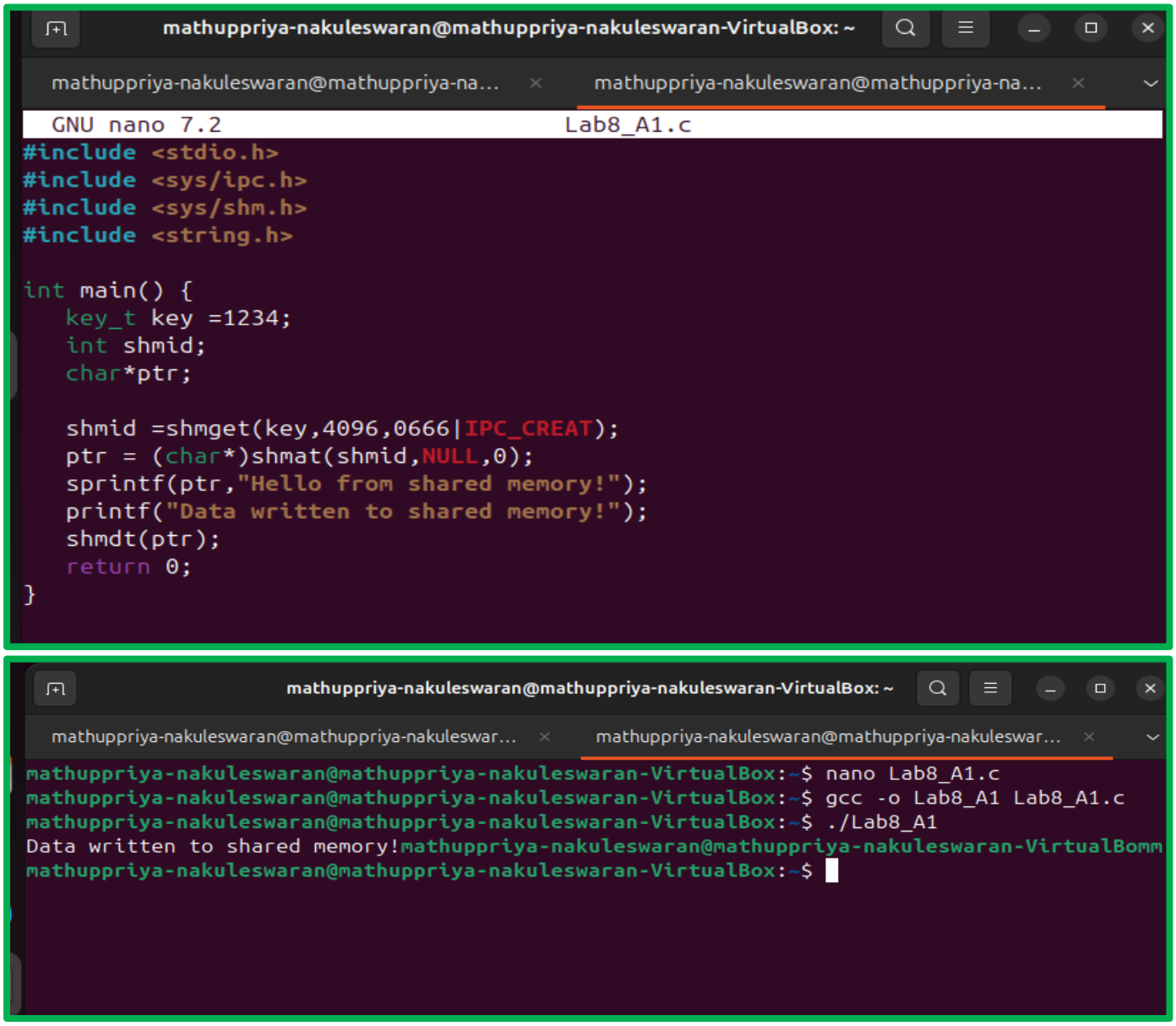


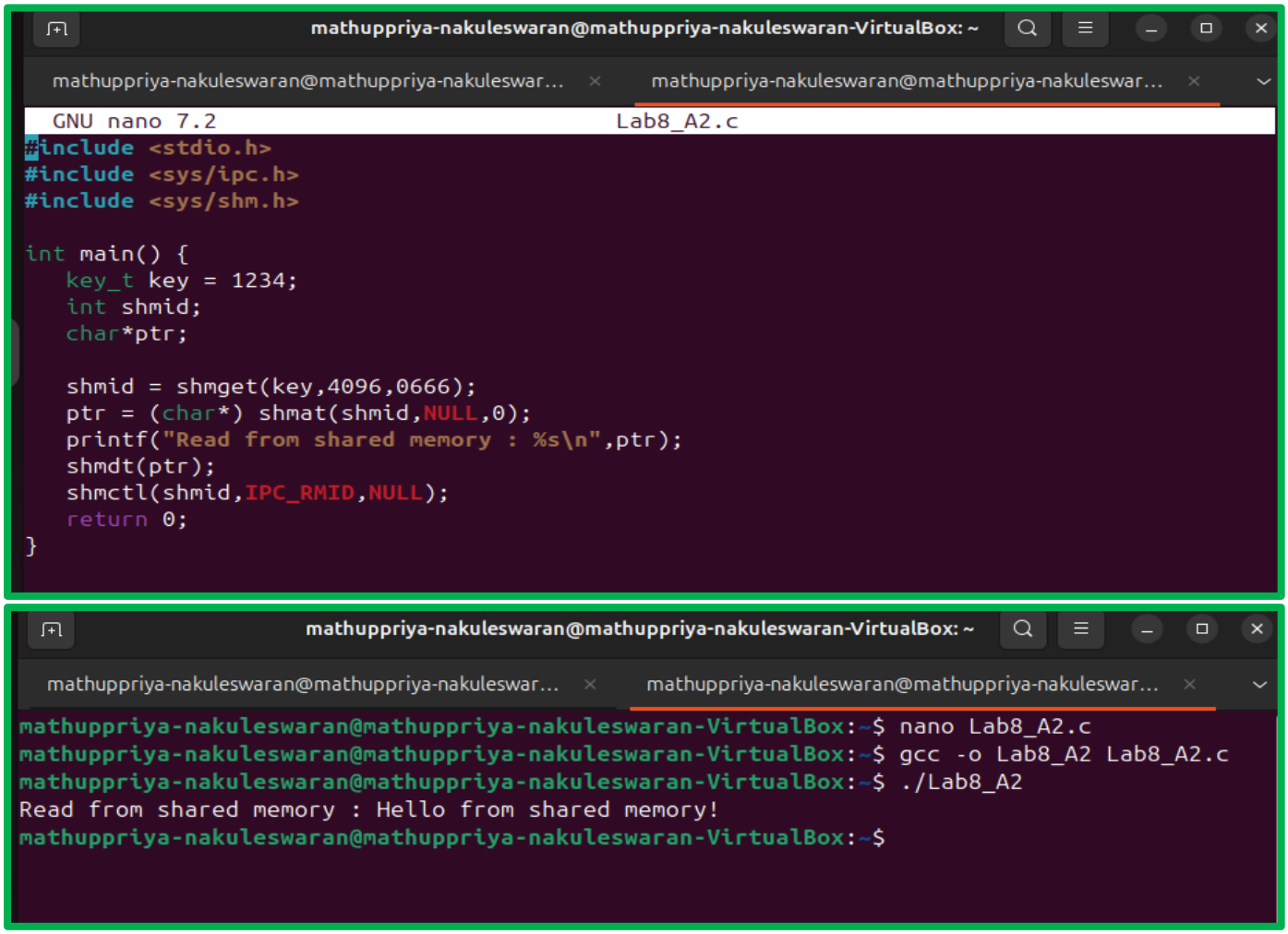
Figure 8.1 : Create shared memory and read & write.

Line No	Code	Explanation
1	#include <stdio.h>	used for **input and output functions** like ▪ printf() ▪ sprintf()
2	#include <sys/ipc.h>	It is needed for **IPC features** such as shared memory keys.

3	#include <sys/shm.h>	It contains the shared memory system calls: ▪ shmget() ▪ shmat() ▪ shmdt() ▪ shmctl()
4	#include <string.h>	This is used for string handling. In this code it is not heavily used, but it is commonly included when working with strings.
5	int main() {	The program starts here.
6	key_t key = 1234;	It creates a **key** for the shared memory. To identify the shared memory.
7	int shmid;	This variable stores the **shared memory ID**. After creating or finding shared memory, the system gives back an ID. That ID is stored in shmid.
8	char *ptr;	This is a **pointer to char**. It will point to the shared memory after attaching it. Since the data is text, char * is used.
9	shmid = shmget(key, 4096, 0666 | IPC_CREAT);	**shmget() : get shared memory** ▪ It either: creates a new shared memory segment, or gets an existing one. **Parameters:** key → identifies the shared memory 4096 → size in bytes 0666 → read/write permission IPC_CREAT → create the memory segment if it does not already exist. The shared memory segment is created, and its ID is stored in shmid.

10	ptr = (char *) shmat(shmid, NULL, 0);	This attaches the shared memory to the process **shmat() : shared memory attach (**Now the process can use the memory.) **Parameters:** shmid → shared memory ID NULL → let system choose the address 0 → normal access mode ptr now points to the shared memory block. So now, ptr can be used like a normal string variable in memory.
11	sprintf(ptr, "Hello from shared memory!");	This writes the message into shared memory. **sprintf() :** write formatted data into a string Here: destination = ptr text = "Hello from shared memory!" So this line stores that message in the shared memory.
12	printf("Data written to shared memory\n");	This prints a normal output message on the screen. Here It tells the user that data was written successfully.
13	shmdt(ptr);	This detaches the shared memory from the process. **shmdt() : shared memory detach** The program is done using the shared memory, so it disconnects from it. This does **not delete** the shared memory. It only removes this process's connection to it
14.	return 0;	This ends the program successfully.

Activity 2



Line No	Code	Explanation
1	#include <stdio.h>	need for printing output
2 3	#include <sys/ipc.h> #include <sys/shm.h>	need for IPC need for shared memory functions
4	int main() {	Program starts here.
5	key_t key = 1234;	▪ Same key as the first program. ▪ important because the reader program must use the **same key** to access the same shared memory.
6	int shmid;	shmid stores shared memory ID.
7	char *ptr;	ptr points to the attached shared memory
8	shmid = shmget(key, 4096, 0666);	This gets the existing shared memory segment.

		▪ Here there is **no IPC_CREAT :** Because this program is only trying to **access existing shared memory**, not create a new one. ▪ If the memory is already created by the first program, this line gets its ID.
9	ptr = (char *) shmat(shmid, NULL, 0);	Attach the shared memory to this process. Now ptr points to the shared memory where the message is stored.
10	printf("Read from shared memory: %s\n", ptr);	This prints the contents stored in shared memory. Since ptr points to the message, the output will be something like: Read from shared memory: Hello from shared memory!
11	shmdt(ptr);	Detach the shared memory from this process.
12	shmctl(shmid, IPC_RMID, NULL);	This line removes the shared memory segment. **shmctl() :** control operation on shared memory **Parameters:** shmid → shared memory ID IPC_RMID → remove the shared memory NULL → no extra structure needed So this line **deletes** the shared memory from the system. Important difference: shmdt(ptr) → detach only shmctl(..., IPC_RMID, NULL) → completely remove shared memory
13	return 0;	End of program.

Activity 3

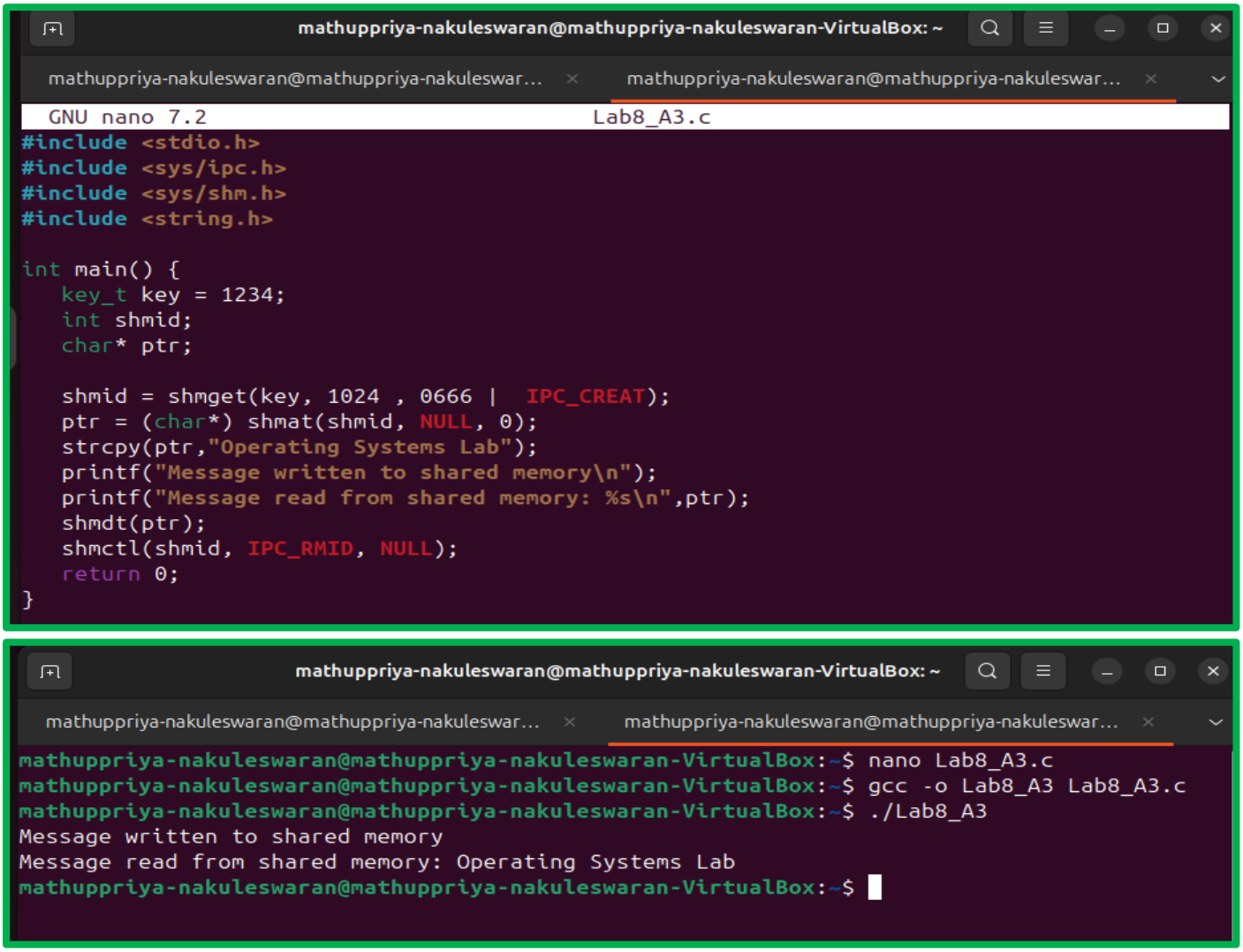


Figure 8.3 : To implement a shared memory using system calls & Create shared memory & write a message into it, Then read the message from the shared memory & print it.